USAF ACADEMY
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
LESSON 29 LAB - Range-Doppler and MTI
Lab Packet
Fall 2026

Name: _____ **Section:** _____ **Date:** _____

Learning Objective 4.5: I can process FMCW radar data to produce range, range-waterfall, and range-Doppler results.

Learning Objective 4.6: I can apply moving target indication (MTI) processing to distinguish moving targets from clutter.

This packet is the L29 Lab, in three parts, and you turn it in once, complete, with your notebook, before Lesson 30. **Part 1, Predict**, is yours to do before the lab period: the shaded Predicted cells of each table. **Part 2, Measure**, is your own kit's readings and your notebook's in the white Measured cells of the same tables. **Part 3, Analyze**, is yours to write after the lab.

Grading. One submission, counted toward your engagement grade and graded as a complete, documented attempt. I grade each Part 3 answer on four things:

- a claim: what agrees, what does not, and why;
- the evidence: your own numbers from Part 2, set against your predictions;
- the reasoning that connects the evidence to the claim;
- the uncertainty: how large a difference your measurement can resolve.

Part 1: Predict (individual, before the lab period)

1. **Predict.** Fill in every shaded **Predicted** cell of Tables A to F. The lesson's Pre-Lab frame names the frame that works each method, on a teaching burst; work yours at the kit's numbers below. Attach your work for one row of each table on your own paper. Use $c = 3.00 \times 10^8$ m/s; velocity is positive closing.

    - **The burst.** The Phaser app's **Range + Velocity** preset: a 9.9 GHz carrier, chirps of 500 MHz in 300 μs sampled at 2 MSPS, and 256 chirps, one every 0.5 ms. Your 0 m frequency is the one you measured in the L28 Lab.
    - **The scene.** A wall stands 2.50 m from the array by the tape. A push closes at 1.0 m/s, and a pull opens at 1.5 m/s. A hand held still 0.8 m out has a tremor of 2 mm either way at 0.5 Hz, and a hand moved across the beam at that range swings 0.20 m either way once a second. The floor row of Table E is read at 14 m/s. Predict as if the chirps were timed perfectly.
    - **The spinner.** Three identical arms turn on a 3.0 cm radius at 15 revolutions a second. Edge-on, its plane of rotation holds the line of sight; face-on, a hand holds it 10° off face-on.
    - **The notebook.** Table D's shorter bursts are the first M chirps of the 256, and its 200 MHz burst is the preset with that bandwidth. Table F's windows are N_{win} chirps, each a hop of $N_{\text{win}}/4$ after the last.

Part 2: Measure (your kit in class, the notebook after)

Follow the steps on the L29 Lab page and fill in every white **Measured** cell. Read the app's levels by hovering over the Range-Doppler view; the notebook's levels are in dBFS, on a scale about 18 dB above the app's, because the app uses a Blackman window and scales its FFT differently from `range_doppler`, which uses a Hann window and divides by the window's sum. Compare changes, not levels, between the two. Mark any reading taken from the course's recorded data with an R.

2. **Table A: the static scene** (Steps 2 and 4, the app).

Quantity	**Predicted**	**Measured**
0 m Beat, f_0		
Leakage: range, velocity		
Leakage level (dB)	–	
Wall: tape range	–	
Wall: range, velocity on the map		
Wall level (dB)	–	
Static-scene file	–	

3. **Table B: a hand** (Steps 5 and 6, the app). Convert each measured velocity to its Doppler frequency.

	Range	**Velocity**		**Doppler**	
	Measured	**Predicted**	**Measured**	**Predicted**	**Measured**
Push					
Pull					
Push file					

4. **Table C: a fidget spinner** (Steps 7 to 9, the app).

Quantity	**Predicted**	**Measured**
Arm radius a, ruler	–	
Range of the band	–	
Band edges, edge-on		
Spacing of the lines		
Spin rate from the edge	–	
Spin rate from the lines	–	
Band edges, face-on		
Spinner file	–	

5. **Table D: chirp count and bandwidth** (Steps 14 and 15, the notebook). The cell is Δv for the chirp counts and ΔR for the 200 MHz burst, and the width is the width of the hand's spot 3 dB down, along the same axis. Record the floor at 256 chirps in dBFS, and at 128 and 64 its change from that.

	Cell		Width	Floor	
Burst	**Predicted**	**Measured**	**Measured**	**Predicted**	**Measured**
256 chirps				reference	
128 chirps					
64 chirps					
200 MHz		–		–	–

6. **Table E: MTI** (Steps 10 to 12, the app, and Step 16, the notebook). Record the level with MTI off, and for each canceller the change from that level in dB, or "gone" where the return drops to the noise floor.

	Off	2-pulse		3-pulse	
Return	**Measured**	**Predicted**	**Measured**	**Predicted**	**Measured**
Wall, app					
Pushing hand, app					
Hand held still, app					
Hand across, app					
Your hand, notebook	–				
Floor near 14 m/s, notebook	–				

7. **Table F: the spinner's spectrogram** (Step 17, the notebook). Your `R_SPIN` _____ m.

	Doppler cell		Column step		Band, 10 dB	
N_{win}	**Predicted**	**Measured**	**Predicted**	**Measured**	**Predicted**	**Measured**
64						
32						

Part 3: Analyze (individual, after the lab)

Answer each prompt from your own Part 2 readings and your own Part 1 predictions.

8. **The chirp count.** Compare your Table D readings at 256, 128, and 64 chirps with your predictions. State what agrees and what does not, give the cause you think most likely for each difference, and say how large a difference your readings can resolve.

9. **The spinner.** Find your spinner's spin rate every way your Table C allows. Compare the results with each other and with your prediction, and say which you would report, with what uncertainty, and why.

10. **The cancellers.** Compare what each canceller did to each return in your Table E with your predictions, on the app and in the notebook. State where they agree and where they do not, give the cause you think most likely for each difference, and describe a measurement on this kit that would test it.

Documentation: _____